

Curriculum Vitae of Federico Zocco

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PERSONAL INFORMATION

Given name: Federico

Family name: Zocco

Address: [SIRSLab](#), Department of Information Engineering and Mathematics, University of Siena, Via Roma 56, Siena, Italy

Email: federico.zocco.fz@gmail.com

Year of birth: 1990

Nationality: Italian

Personal homepage: <https://fedezocco.github.io/>

Full list of publications and preprints: [Google Scholar profile](#)

Open-source codes: <https://github.com/fedezocco> and <https://github.com/cirokearch>

RESEARCH STATEMENT

I am a researcher and engineer specializing in *circular intelligence, robotics, and control*. My research is strongly oriented towards the actual implementation of a circular economy and it combines deep-learning vision, control systems, compartmental thermodynamics, robotics, and network science. To do so, I leverage my exceptionally multidisciplinary background ranging from Mechanical Engineering (BSc) to Robotics (MSc), Machine Learning (PhD), and Circular Economy (post-doc). Please refer to my personal homepage for details: <https://fedezocco.github.io/>.

SELECTED PAPERS

1. **Zocco, F.**, Lin, T.C., Huang, C.I., Wang, H.C., Khyam, M.O. and Van, M., 2023. Towards more efficient EfficientDets and real-time marine debris detection. *IEEE Robotics and Automation Letters*, 8(4), pp. 2134-2141.
2. **Zocco, F.**, Sleath, D. and Rahimifard, S., 2025. Towards a thermodynamical deep-learning-vision-based flexible robotic cell for circular healthcare. *Circular Economy and Sustainability*, pp.1-23.
3. **Zocco, F.**, Sopasakis, P., Smyth, B. and Haddad, W.M., 2023. Thermodynamical material networks for modeling, planning, and control of circular material flows. *International Journal of Sustainable Engineering*, 16(1), pp.1-14.
4. **Zocco, F.**, Lake, D.R., McLoone, S. and Rahimifard, S., 2025. Synchronized object detection for autonomous sorting, mapping, and quantification of materials in circular healthcare. *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1-10.
5. **Zocco, F.**, Corti, A. and Malvezzi, M., 2025. CiRL: Open-source environments for reinforcement learning in circular economy and net zero. *arXiv preprint arXiv:2505.21536*.
6. **Zocco, F.**, Dragusanu, M. and Malvezzi, M., 2025. A biofueled renewable minimally-critical biodegradable easy-to-disassemble robotic manipulator: Initial design. *TechRxiv preprint techrxiv.176704884.43305930*.

PROFESSIONAL EXPERIENCE

03/2025-Present	<i>Postdoctoral researcher</i> at the Siena Robotics and Systems Lab (SIRSLab), University of Siena, Italy Line manager: Prof. Monica Malvezzi
05/2024-02/2025	<i>Postdoctoral researcher</i> based in Loughborough, England, UK
12/2022-05/2024	<i>Research associate</i> of the ReMed project, Loughborough University, England, UK Line manager: Prof. Shahin Rahimifard
10/2021-05/2022	<i>Research fellow</i> at the i-AMS centre, Queen's University Belfast, Northern Ireland, UK Line manager: Prof. Mien Van
02/2021-05/2021	<i>Research engineer</i> , Queen's University Belfast, Northern Ireland, UK Line manager: Prof. David Laverty
10/2017-12/2018	<i>Teaching assistant</i> , Queen's University Belfast, Northern Ireland, UK
20/11/2017-30/11/2017	<i>Volunteer teacher</i> at the orphanage "The Potter's Village" in Dodowa, Accra, Ghana, with the voluntary organization "Soho – Sparks of Happiness Onlus" based in Florence, Italy
03/2016-10/2016	<i>Research student</i> , EPIC research cluster, Queen's University Belfast, Northern Ireland, UK Advisors: Prof. Alberto Landi, Prof. Seán McLoone, Prof. Xueqin Liu, and Prof. Emanuele Crisostomi Activity: Development of the MSc thesis

03/2013-09/2013	<i>Research student</i> at the E. Piaggio research center, University of Pisa, Italy Advisors: Prof. Antonio Bicchi, Dr. Giorgio Grioli, and Dr. Manuel G. Catalano Activity: Development of the BSc thesis
2011-2016	<i>Occasional ski instructor</i> (worked with three Italian schools/teams: “Scuola sci e snowboard Cristallo” in Cortina d’Ampezzo, “Scuola sci Monte Amiata” in Siena and “Sci club Arcidosso” in Grosseto)

EDUCATION

01/2017-12/2021	<i>Ph.D. in Applied Machine Learning, Queen’s University Belfast, Northern Ireland, UK</i> Thesis title: Advances in machine learning for sustainable manufacturing Advisor: Prof. Seán McLoone
10/2013-11/2016	<i>Master of Science degree in Robotics and Automation Engineering, University of Pisa, Italy</i> Grade: 110/110
10/2009-10/2013	<i>Bachelor of Science degree in Mechanical Engineering, University of Pisa, Italy</i>

AWARDS

- Feb 2026: Seal of Excellence awarded in the 2025 MSCA Postdoctoral Fellowship call for the project “Toward robotics and AI adoption in circular economy: Application to WEEE disassembly”
- 01/01/2017-31/12/2020: Full Ph.D. scholarship awarded by Irish Manufacturing Research
- 01/02/2016: Studentship awarded by University of Pisa to support students with the highest marks visiting foreign institutions (1500 €) (I visited Queen’s University Belfast, UK)

SUPERVISION

Co-supervised 4 Ph.D. students, 3 BSc/MSc students, and coached people of various ages (3-60 years old) and skill levels (from beginners to athletes) when working as a ski instructor (period 2011-2016).

PERSONAL SKILLS

Languages: English, Italian

Programming languages: MATLAB, Python, C++, C

Engineering software: Simulink, Wolfram Mathematica, SolidWorks, ProEngineer, REFPROP, DIgSILENT, Android Studio, RoboDK, Autodesk Fusion

Deep learning frameworks: PyTorch, TensorFlow/Keras

Reinforcement learning libraries: Stable Baselines3 (for learning algorithms), Gymnasium-Robotics (for environments), panda-gym (for environments)

Hardware: Nvidia Jetson Nano, STM32F4DISCOVERY, Arduino Uno, Raspberry Pi; assembled a BlueROV2; programmed an ABB IRB 120

Sport skills: Diploma of ski instructor obtained in 2009; alpine ski athlete in the period 2004-2008 ([results](#)).